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PROJECT AND TEAM INFORMATION

Project Title

VidOptix - Video Compression System.

Team Name: *Team #*	*BitSquad DSCPP-III-2026-T198*
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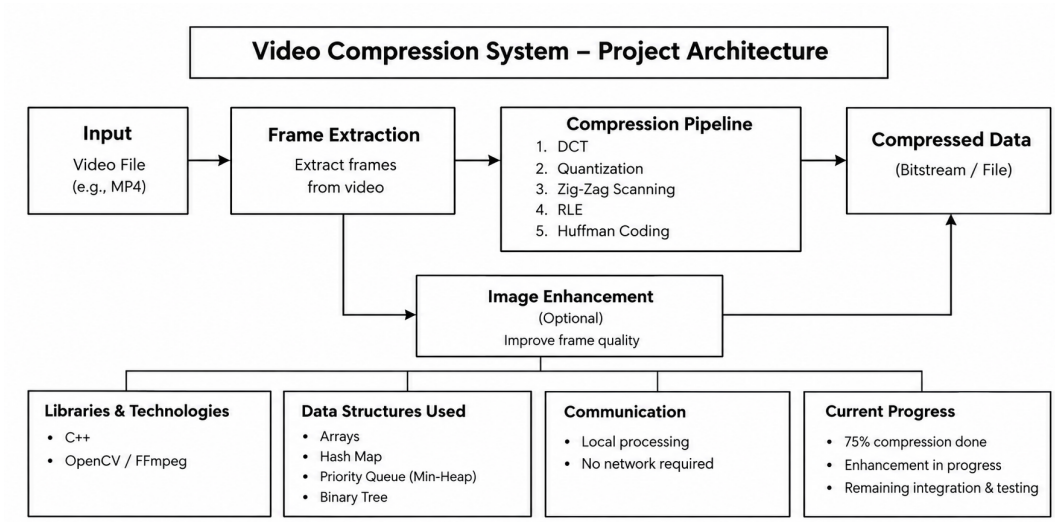
PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

Project Abstract
This project is about creating a video compression system. The goal is to make video files smaller. The videos should still look good. The system works on each video frame. It uses methods like Discrete Cosine Transform. This method helps to change the way video data is stored. Another method is quantization. This step removes some details that're not very important. Then there is zig-zag scanning. This helps to organize the data in a way. Run-Length Encoding is used next. This method makes the data shorter. The last step is Huffman coding. This is a way to encode the data so that it takes space.
The compression process changes the frame data into frequency coefficients. Then it removes some of the important information. The data is then turned into a form. We are also trying to improve the way videos look. These image enhancement techniques help to make the video look better.
Our team has done 75 percent of the compression part. We have started working on the enhancement techniques. The next steps are to finish the compression system. We also need to add the enhancement features. Then we will test how well everything works.

Updated Project Approach and Architecture (2 pts)

1. System Design: Take video frames and divide them into 8, by 8 blocks then apply DCT, quantization, zigzag scanning and run-length encoding to make the video smaller.
2. Algorithms and Data Structures: Use Huffman coding with a min heap priority queue and a binary tree to compress data in the way possible.
3. Libraries and Technologies: Write the code using C++. Use OpenCV or FFmpeg to work with video files.
4. Communication Protocols: Handle video files without needing any network communication.
5. Progress: Our team has completed seventy-five percent of the compression work and has started looking at ways to improve image quality..
6.Testing: Finish the remaining modules and measure file size reduction, processing time and the visual quality of the output.



Tasks Completed (7 pts)

Task Completed	Team Member
1. Input and Frame reading	Yaduveer Yadav
2. Frame Extraction	Yaduveer Yadav , Anshika Goswami
3. DCT and 8X8 block division	Yaduveer Yadav , Deepanshi Chaudhary
4. Enhancement model	Deepanshi Chaudhary , Mohd Ubaid
5. UI and Frontend model	Anshika Goswami , Mohd Ubaid

Challenges/Roadblocks (7 pts)

Challenges/Roadblocks

1. Maintaining Visual Quality: The challenge is to cut file size while keeping video look good. One way is to tweak quantization levels.
2. Algorithm Integration: The task is to merge DCT, quantization, zig-zag scanning, RLE and Huffman coding. All steps must fit into one pipeline.
3. Data Structure Implementation: We need to use hash maps, priority queues and binary trees for Huffman coding.
4. Image Enhancement: The goal is to make frames look better but not to slow down the process or hurt compression.
5. Time Management: We need to finish the rest of the compression modules and enhancement features before the deadline. This means splitting tasks and integrating often.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Main Tasks Remaining 1. Complete Compression Implementation 2. Image Enhancement 3. Backend and Frontend connection	TEAM

Project Outcome/Deliverables (2 pts)

1. Working Video Compression System: A system built in C++ that reduces the size of video files by using DCT, quantization, zig-zag scanning, RLE and Huffman coding.
2. Data Structure Implementation: Real-world application of arrays, hash maps, priority queues (min-heaps) and binary trees during the compression steps.
3. Image Enhancement Features: Techniques applied to enhance the look of video frames making them clearer and more visually appealing.
4. Performance. Report: A report that shows how much space is saved. It also shows how fast the compression runs. It also shows how well the visual quality holds up. This includes project documentation.

Progress Overview

Progress: three fourths of the compression system has been finished.
Ahead of Schedule: The user interface development is happening faster than we had expected.
On Schedule: Frame reading, frame extraction, DCT implementation and dividing frames, into 8x8 blocks are all going as planned.
Remaining Work: Complete the remaining compression parts include image enhancement methods and do testing and performance evaluation.

Codebase Information (2 pts)

- Repository Link: https://github.com/yaduveery41/pbl$_{v}idoptix_{3}rd_{s}emesterpbl_Vidoptix_3rd_semester$

- Branch: Phase-2

- Important Commits: Our team has committed each individual development step to the repository, maintaining a record of the implementation progress and changes made throughout the project.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
1. Frame Extraction	PASS	frame extraction successful
2. 8x8 Block DCT implementation	PASS	DCT implementation successful
3. Quantization and RLE	PASS	Quantization and RLE implemented successfully
4. 50 percent Enhancement	PASS	
5. UI and Frontend	PASS	UI and Frontend verified and functional

Deliverables Progress (2 pts)

- Video Compression System. In Progress: Video Compression System is three quarters finished. Video Compression System has read frames taken parts out split each frame into 8×8 squares and performed DCT on those squares.
- Data Structure Implementation. In Progress: Data Structure Implementation uses arrays, priority queues and binary trees while Video Compression System works. Data Structure Implementation helps keep data organized during compression.
- Image Enhancement Features. In Progress: I am studying Image Enhancement Features to make frames look better. Image Enhancement Features will give pictures after compression.. Performance. Documentation. Pending: Performance Evaluation and Documentation still needs tests to measure how much data is saved, to check how long compression takes to judge quality and to write up the whole project.